Aerospace Engineering Graduate Programs — Application Guide

For Olivia — applying with a theoretical physics master's from NBI, targeting Fall 2027 admission.

How AE graduate admissions work

AE admissions in the US are broadly similar to physics PhD programs — you apply to the department, not to a specific advisor, though identifying faculty you want to work with is important and expected. A few differences:

- Master's vs. PhD: In the US, most funded positions are PhD programs. Standalone master's programs often require you to pay tuition or are unfunded. A funded PhD is always preferable if you're committed to research. Some students enter as MS and transition to PhD; others enter directly into PhD programs. Clarify with each program.
- Engineering GRE: Some programs still request a general GRE. An engineering-specific GRE is rare now. Check each program's current policy.
- Letters of recommendation: Three, ideally from research supervisors. Obers and Holger Bech Nielsen at NBI are strong. Your CERN supervisor counts.
- Statement of Purpose: More important than the GRE. Should articulate what sub-area you want to work in and why — and name specific faculty. Be honest about your physics background and direct about where you want to take it.

Your profile — how to present it

You are not a typical AE applicant. That is an asset if framed correctly.

What works for you:

- NBI master's in theoretical physics — signals mathematical maturity and serious research capability
- CERN research experience — institutional name recognition, concrete research credential
- Bachelor's thesis on string theory and black holes — demonstrates ability to engage with hard theoretical problems
- Strong mathematical background — directly applicable to electric propulsion, astrodynamics, space environment

What to address:

- No prior AE or engineering coursework — acknowledge it directly in the Statement of Purpose, then pivot to why your background is relevant to the specific sub-area you're targeting. "I come from a plasma physics background, which I believe is directly

applicable to Hall thruster research" is more effective than silence on the gap.
- Be specific about sub-area — "aerospace engineering" is too broad. "Electric propulsion, specifically plasma instabilities in Hall thrusters" or "astrodynamics for cislunar mission design" tells the committee where to route your application and who to show it to.

Timeline for Fall 2027 admission

When	What
Summer 2026 (now)	Research programs and faculty; identify 2–3 target faculty per school
August–October 2026	Email faculty; introduce yourself and your background
September 2026	Ask recommenders (Obers, Holger Bech Nielsen, CERN supervisor)
October–November 2026	Draft Statement of Purpose
December 2026–January 2027	Most US AE deadlines fall here; submit early
February–March 2027	Interview invitations and visit days
April 15, 2027	Decision deadline — same Council of Graduate Schools agreement as physics

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Contacting faculty

Do this — same advice as for physics programs, same timing.

For AE, the research hook is even more important because your background is non-standard. Faculty want to know you've read their work and understand what they actually do. A specific email from someone with a plasma physics background asking a genuine question about Hall thruster instabilities will get read. A generic email from a physicist saying they're interested in "aerospace" will not.

Template: same structure as the physics email templates. The key adaptation — be explicit about your physics background and why it connects to their specific research rather than the field generally.

The background gap — practical steps

If you want to strengthen the engineering side before applications or during a master's:

- Online coursework: MIT OpenCourseWare has full aerodynamics, thermodynamics, and orbital mechanics courses. Working through one or two before or during applications shows initiative and fills vocabulary gaps.
- arXiv and AIAA papers: Read papers from the groups you're targeting. Understanding the

language and open problems before you email faculty makes the email substantively better.
- Talk to Obers: He may know people in the AE community through interdisciplinary connections, particularly in space plasma physics. Ask.

International programs — key differences

TU Delft (Netherlands): Apply through the TU Delft Graduate School. Programs in English. EU tuition (~€2,200/year). Competitive but less so than top US programs. Excellent option.

ISAE-SUPAERO (France): Apply directly. Programs increasingly in English. EU tuition. Research master's available. Strong industry ties.

KTH (Sweden): EU tuition (free for EU citizens). Apply through the KTH admissions portal. Programs in English. Excellent quality of life.

Imperial College (UK): Post-Brexit, Danish students pay international fees (~£32,000/year for engineering). Requires scholarships to be viable. The Schrödinger Scholarship and other external awards are routes.

Money — what to expect and what to assume

The first distinction: stipend vs. scholarship

A stipend comes with a funded PhD position — it is the regular payment that accompanies the research appointment. A scholarship is a separate competitive award you apply for independently. These are not the same thing, and you should not assume one is the other.

US PhD programs

Most funded PhD programs in AE cover tuition and pay a stipend. These are research assistant (RA) or teaching assistant (TA) appointments — employment, not charity. If you are admitted to a funded PhD position, the stipend comes with it automatically.

The critical point: standalone master's programs in US AE are often unfunded. You pay tuition. Some students enter as MS and hope to transition to a funded PhD — this is a real path but carries financial risk. Apply to PhD programs if you want funding.

Approximate stipend ranges for funded PhD positions (not guaranteed until you have an offer):

Program	Stipend/yr	City cost
MIT AeroAstro	~$40–44k	High (Boston)
Princeton MAE	~$40–44k	Moderate (NJ)

Caltech GALCIT	~$38–42k	High (LA)
CU Boulder	~$28–33k	Moderate
Georgia Tech	~$28–32k	Moderate (Atlanta)
UT Austin	~$26–30k	Moderate-high
Michigan AE	~$25–30k	Low (Ann Arbor)
Purdue AAE	~$22–26k	Very low (W. Lafayette)

European programs

At TU Delft and KTH, PhD students are legally employees — the salary is attached to a specific funded position advertised by a supervisor, not to the institution generally. You cannot simply apply to the university and expect to be paid. You need to be selected for a specific funded position, or have a supervisor with grant money who agrees to take you.

If you find a funded position: TU Delft pays ~€2,500–2,800/month; KTH pays ~30,000–35,000 SEK/month. Both cover you as an employee with benefits. But finding the position is the work.

ISAE-SUPAERO research master's funding varies by program and year — do not assume funding; check each program explicitly.

Imperial London: funded positions exist but are competitive. Unfunded, London is very expensive on a student budget.

The Danish comparison

Danish PhD students are employees and earn ~390,000–420,000 DKK/year (~$56–61k) with full benefits. Every option above is a nominal pay cut from that. The trade-off is the institution, the supervisor, and where the degree takes her.

See also: landscape.md for sub-field descriptions, programs.md for program details